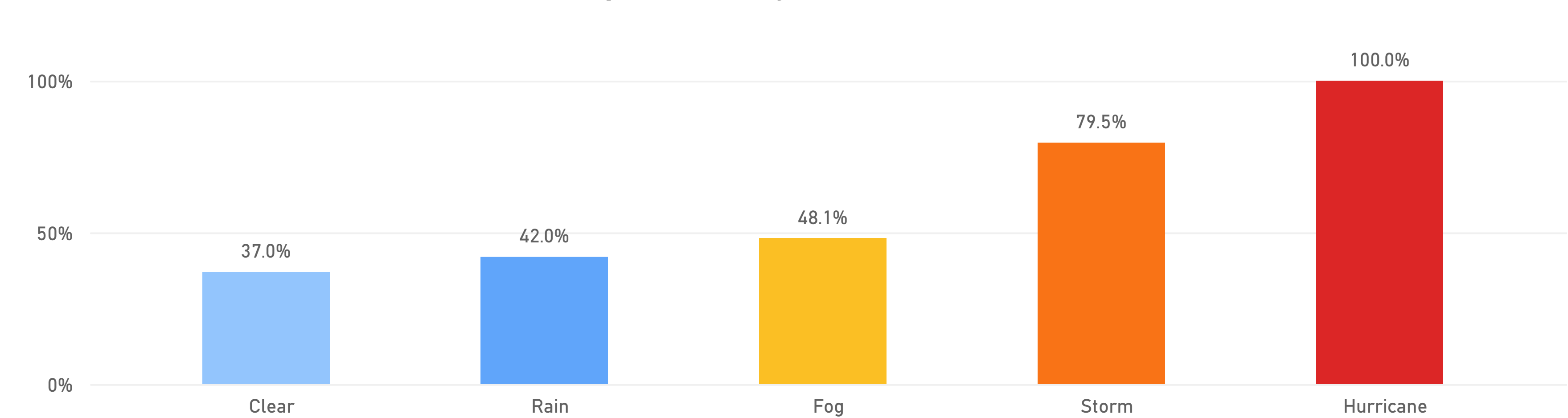


SUPPLY CHAIN RISK OVERVIEW

Simulated shipment data | 2024-2025

Data updated to
September 17, 2026

Disruption Rate by Weather Condition



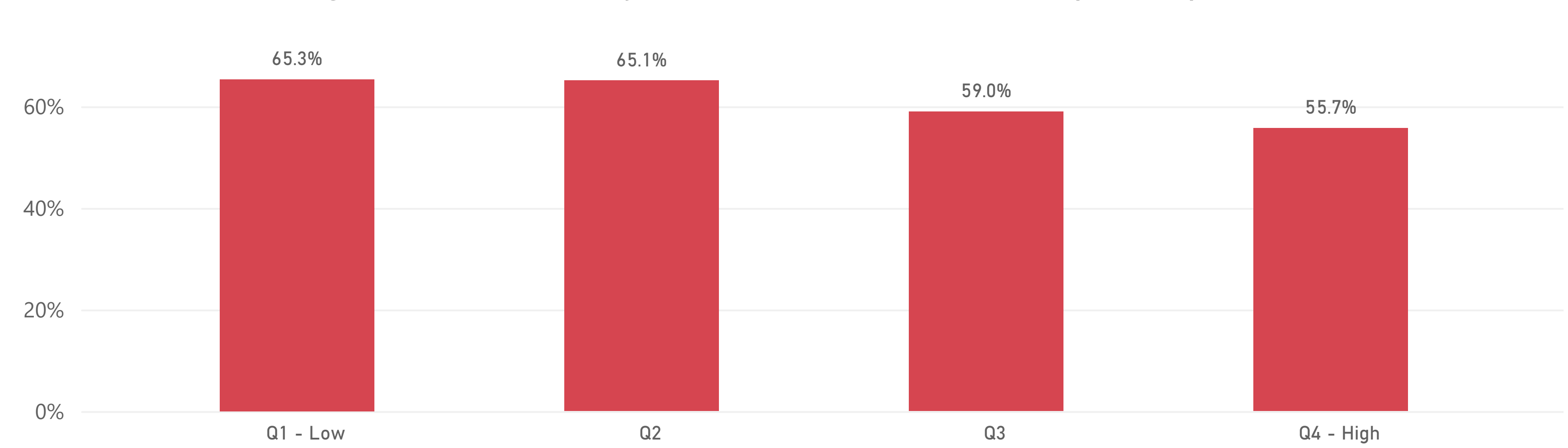
Routes Requiring Further Investigation

High disruption rates help identify where to investigate; they do not imply route-level causality



Disruption Rate by Carrier Reliability

Higher carrier reliability is associated with lower disruption exposure

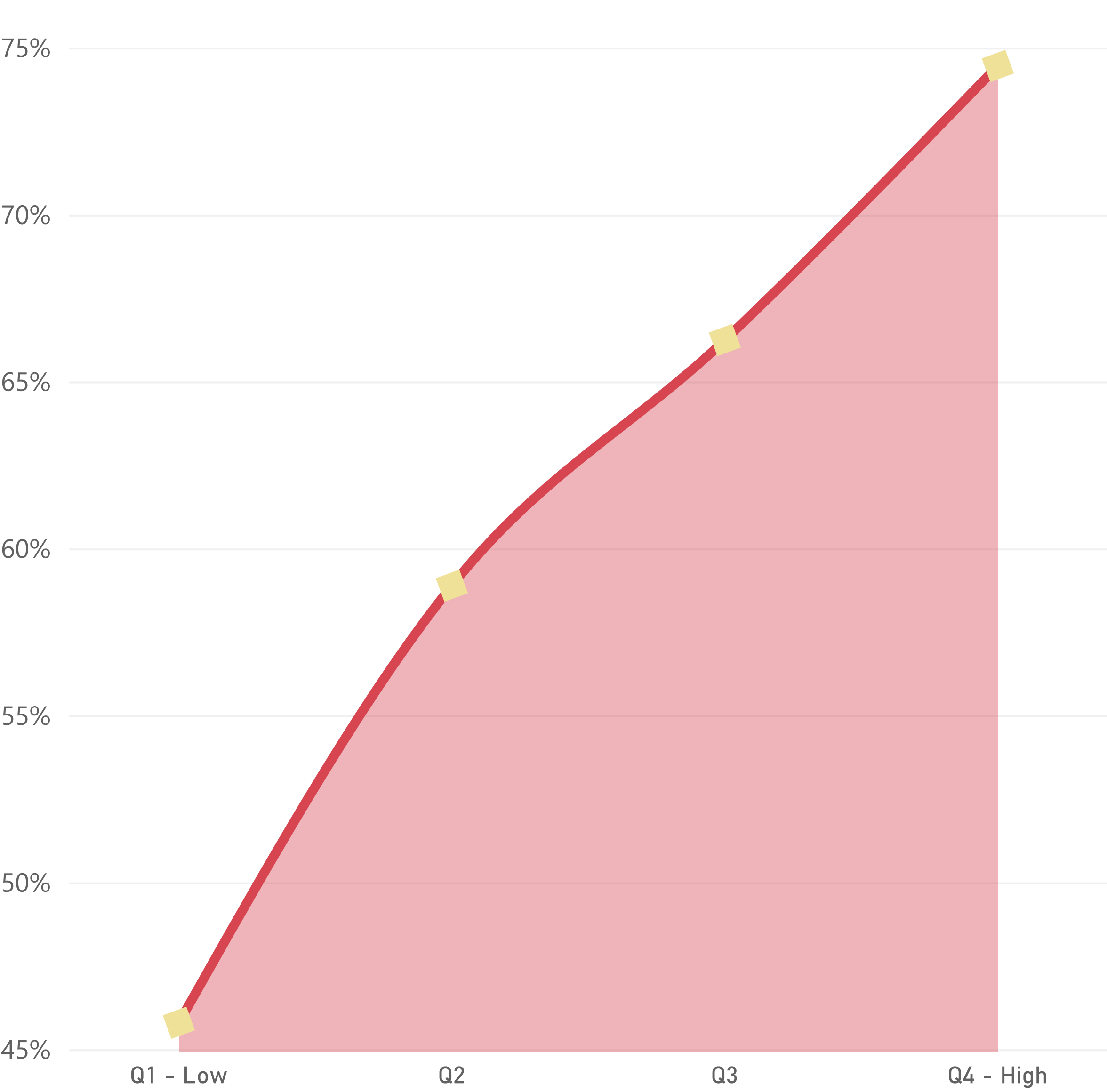


Disruption Risk by Weather & Geopolitical Exposure

Weather_Condition	Q1 - Low	Q2	Q3	Q4 - High	Total
Clear	19.7%	35.9%	42.6%	51.1%	37.0%
Rain	21.3%	37.7%	47.3%	60.2%	42.0%
Fog	29.2%	41.1%	54.6%	67.2%	48.1%
Storm	60.5%	75.8%	87.1%	95.9%	79.5%
Hurricane	100.0%	100.0%	100.0%	100.0%	100.0%
Total	45.8%	58.9%	66.2%	74.5%	61.3%

Disruption Rises with Geopolitical Risk

Risk increases consistently across geopolitical-risk quartiles



Disruption Rate

61.26%

Total Shipments

5,000

Avg Lead Time

19.36

FILTER

Origin_Port

All

Destination_Port

All

Product_Category

All

Weather_Condition

All

Transport_Mode

All

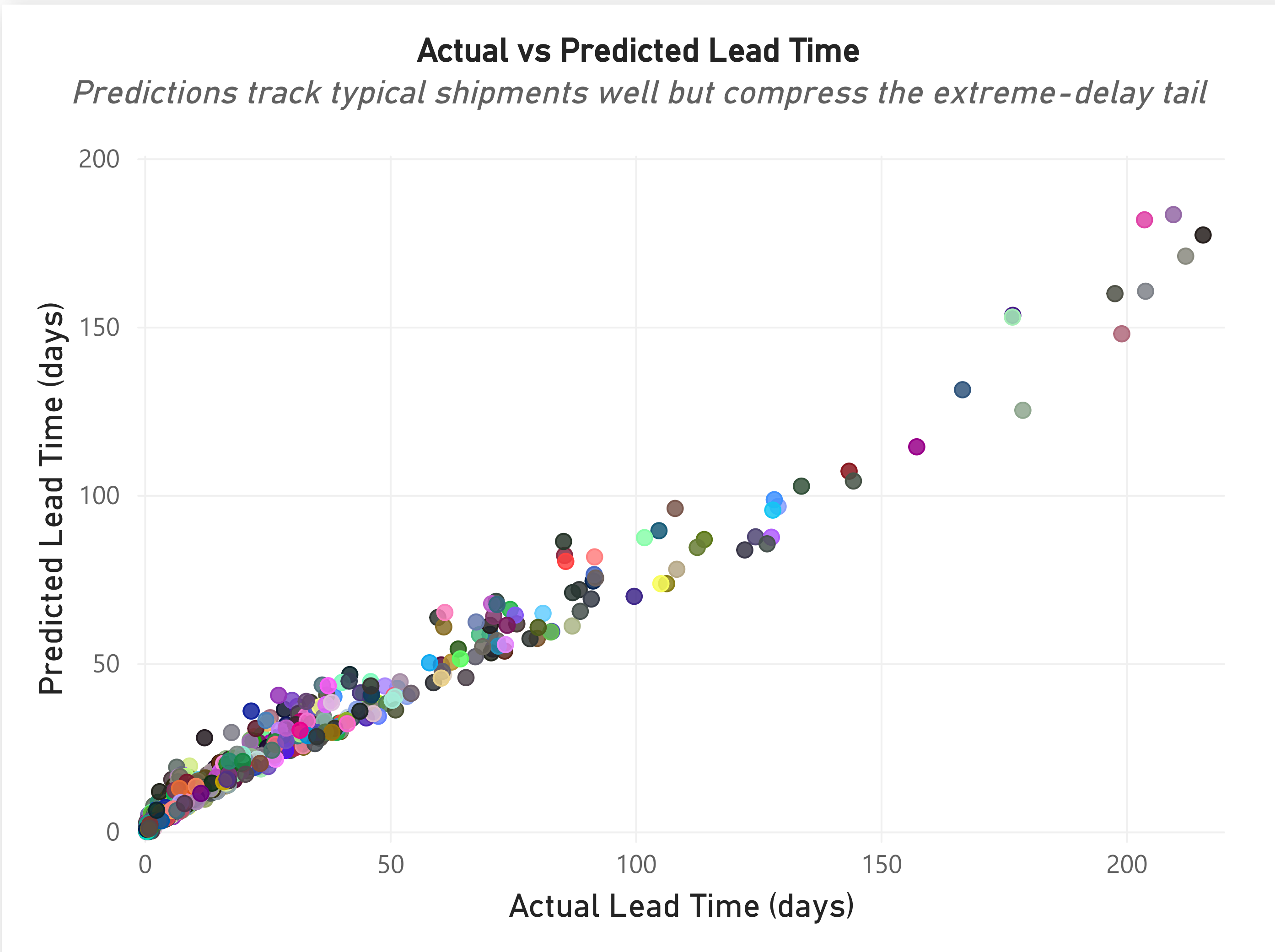
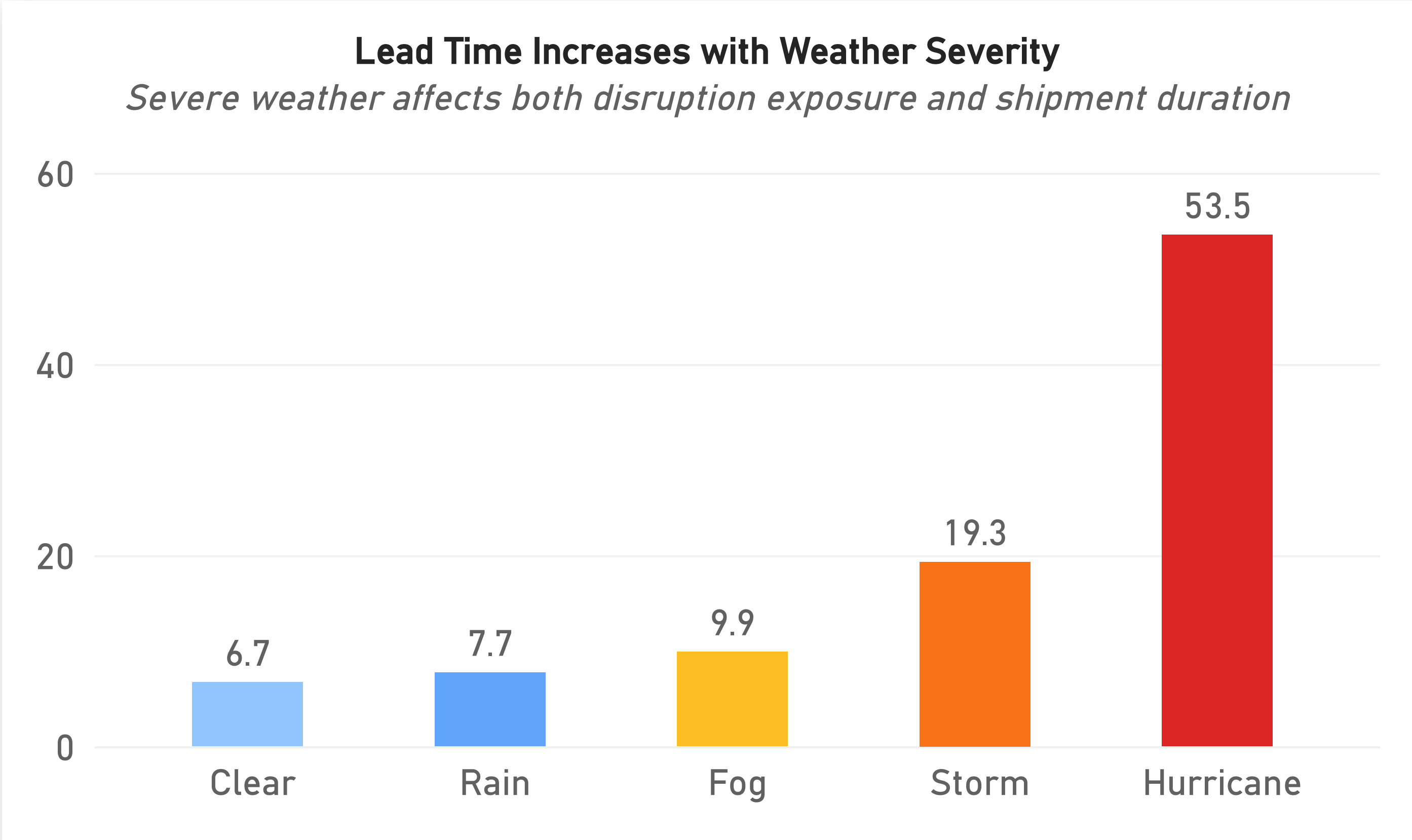
Date

1/1/2024

12/31/2025

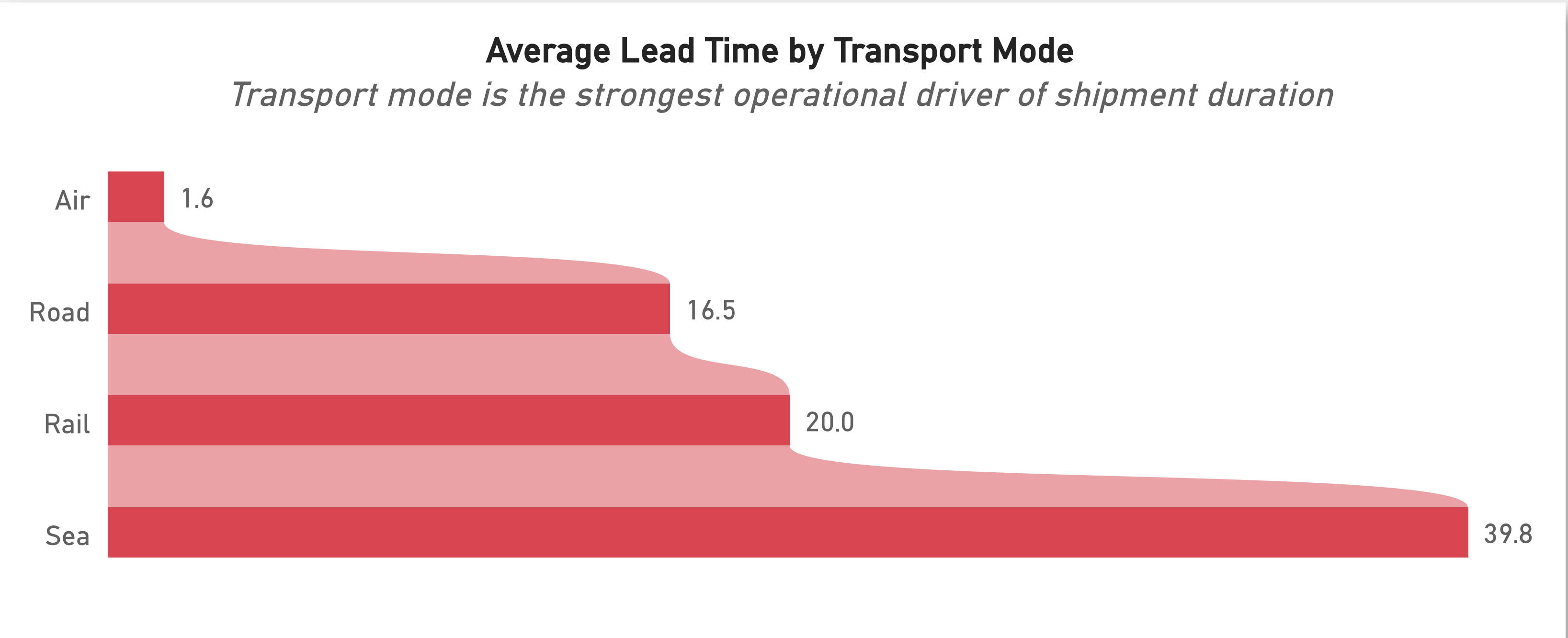
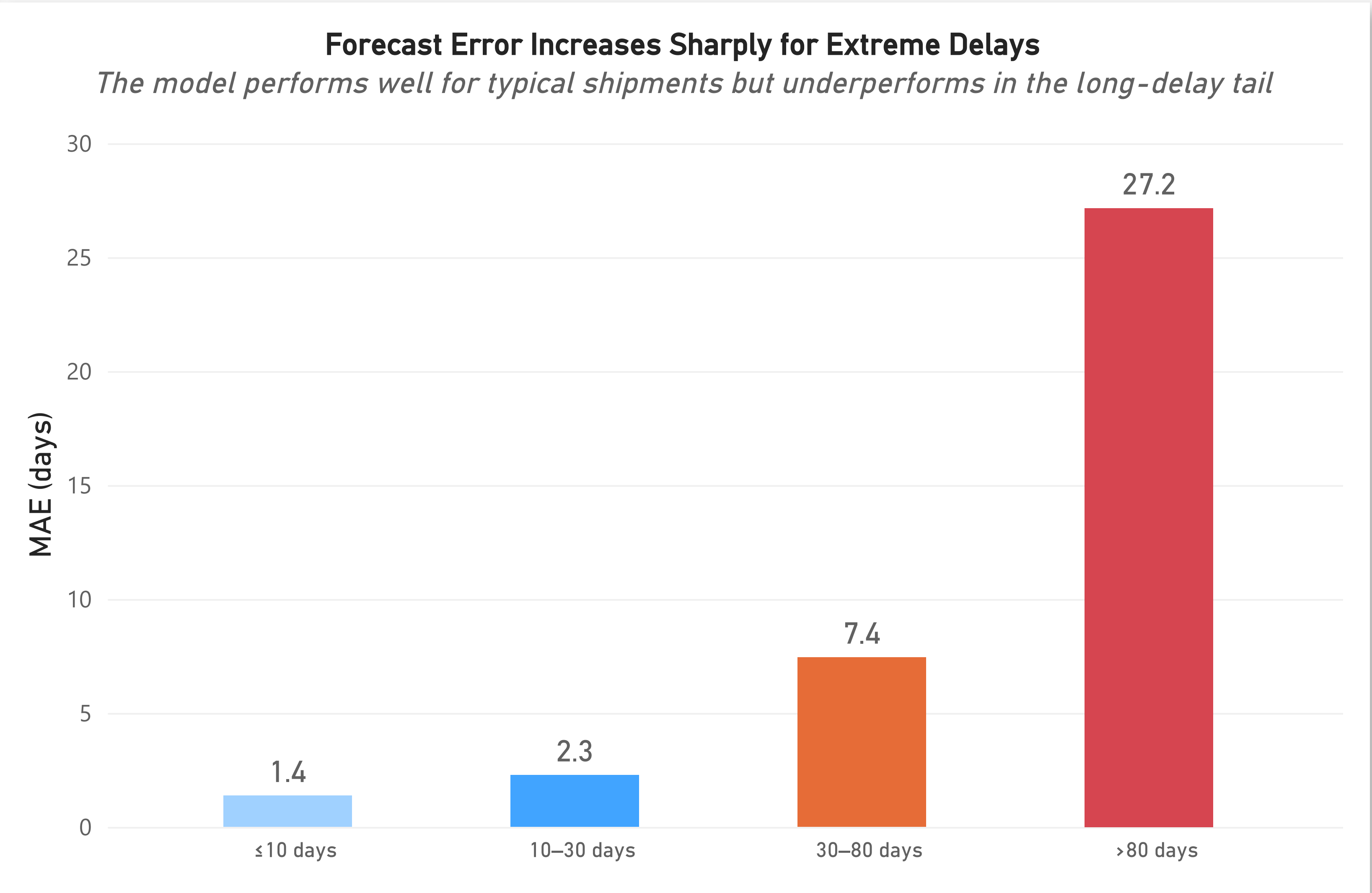
Lead Time & Operational Planning

Simulated shipment data | 2024-2025



Operational Risk

For shipments above 80 days, forecast MAE rises to approximately **27 days**. Extreme-delay shipments should therefore receive additional planning buffers and manual review.



Avg Lead Time
19.36

P95 Lead Time
80.81

Model MAE
3.60

FILTER

Origin_Port

All

Destination_Port

All

Product_Category

All

Weather_Condition

All

Transport_Mode

All

Date

1/1/2024

12/31/2025